Air Quality Forecasting via Gradient Boosting

Course: Semester 5 Applied Machine Learning · **Project 19** · October 2026
Team: Manan Gilhotra, Harshita, Satyam, Yashi Agarwal, Akhil Mishra
Code: github.com/MananGilhotra/air-quality-forecasting

Abstract

We forecast PM2.5 concentration in Beijing **6 hours ahead** from past pollution and weather readings. Using the UCI Beijing PM2.5 dataset (43,824 hourly records, 2010–2014), we engineer 31 features and train an **XGBoost** regressor. We compare it with persistence and seasonal naive baselines in a 4-fold **rolling chronological backtest**. XGBoost achieves an average MAE of **35.8 µg/m³**, against 40.7 for persistence (**12% lower error**), and raises R^2 from 0.52 to 0.65. It wins in every test period. Forecasts are also mapped to US EPA AQI categories, where XGBoost assigns the correct category 53.9% of the time.

1. Introduction and Objective

Fine particulate matter (PM2.5) is one of the most harmful air pollutants. It reaches deep into the lungs and is linked to heart and lung disease. A reliable short-term forecast lets people and city agencies act before pollution peaks, for example by limiting outdoor activity or issuing health warnings.

Objective: forecast hourly PM2.5 at a fixed horizon, and test whether a machine learning model beats simple forecasting rules under a fair, time-respecting evaluation.

Design choice	Our decision
Forecast horizon	6 hours ahead (fixed)
Target	PM2.5 concentration in µg/m³
Main model	XGBoost gradient-boosted trees
Baselines	Persistence, seasonal naive (24 h)
Evaluation	Rolling chronological backtest, MAE, RMSE, R^2
AQI standard	US EPA PM2.5 breakpoints (2024 update)

The gap we address: hand-made lag windows may miss longer time patterns and non-linear interactions between pollution and weather. A gradient-boosted tree model captures non-linear interactions, and a sequence model (LSTM) is planned as the next step for longer patterns.

2. Dataset and Exploratory Analysis

We use the UCI Beijing PM2.5 dataset: hourly readings from the US Embassy in Beijing, 1 Jan 2010 to 31 Dec 2014.

Column	Meaning	Unit
pm2.5	PM2.5 concentration (target)	µg/m³
DEWP, TEMP, PRES	Dew point, temperature, air pressure	°C, °C, hPa
cbwd	Wind direction (NE, NW, SE, calm/variable)	category
Iws, Is, Ir	Cumulated wind speed, hours of snow, hours of rain	m/s, h, h

Size and quality. The file has 43,824 hourly rows. Only PM2.5 has missing values: 2,067 hours (4.7%), mostly in 2010–2012, with the longest gap lasting 155 hours.

Key findings from EDA

- **Skewed target.** The median PM2.5 is 72 µg/m³, the mean is 98.6 and the maximum is 994. Rare severe episodes make RMSE much larger than MAE, so we report both.
- **Strong persistence.** PM2.5 correlates 0.97 with the value 1 hour earlier, 0.75 with 6 hours earlier and 0.40 with 24 hours earlier. This makes "same as now" a strong baseline and motivates lag features.
- **Daily and yearly cycles.** Pollution peaks around midnight (about 113) and dips mid-afternoon (about 86). It is highest in the winter heating months (February about 126) and lowest in late spring and summer (about 80).
- **Wind matters most among weather variables.** Northwest wind brings clean mountain air (median 31 µg/m³), while calm air traps pollution (median 98). Linear correlations with weather are weak (wind speed −0.25), which suggests non-linear effects that suit tree models.

Wind direction	Median PM2.5 (µg/m³)	Hours
NW (northwest)	31	13,484
NE (northeast)	55	4,756
SE (southeast)	91	14,573
Calm / variable	98	8,944

3. Methodology

3.1 Cleaning

We sort by time and resample to a regular hourly interval. Before filling any gaps, we add a `pm25_missing` flag, then forward fill using past values only, so no future information leaks into the inputs. The fully missing first day is dropped. Wind direction is one-hot encoded. The result is 43,800 clean hourly rows, with 2,043 hours flagged as missing.

3.2 Features and target

The **target** is PM2.5 exactly 6 hours after the forecast time. Rows whose future value was originally missing are never used for training or scoring. We build **31 features**, all from current and past hours only:

Group	Features
Current PM2.5	PM2.5 now
Lags	PM2.5 1, 2, 3, 6, 12 and 24 hours ago, and at the same hour yesterday
Rolling means	Average PM2.5 over the last 6 and 24 hours
Calendar	Hour, day of week, month, weekend flag, sin/cos of hour and month
Weather	Dew point, temperature, pressure, wind speed, snow, rain, temperature–dew point gap, 3-hour pressure change
Wind direction	4 one-hot columns (NE, NW, SE, calm)
Missing flag	Whether PM2.5 was measured this hour

3.3 Models

- **Persistence:** forecast = PM2.5 now.
- **Seasonal naive:** forecast = PM2.5 at the same clock hour one day earlier.
- **XGBoost:** 300 gradient-boosted trees (learning rate 0.05, max depth 6, 80% row and column sampling). Each new tree corrects the errors of the trees before it, which captures non-linear pollution–weather interactions. Tree models do not need feature scaling.

3.4 Rolling chronological backtest

Time series must never be shuffled, so we test on the future only. Each fold trains on **all data before** its test period and tests on the **next 6 months**:

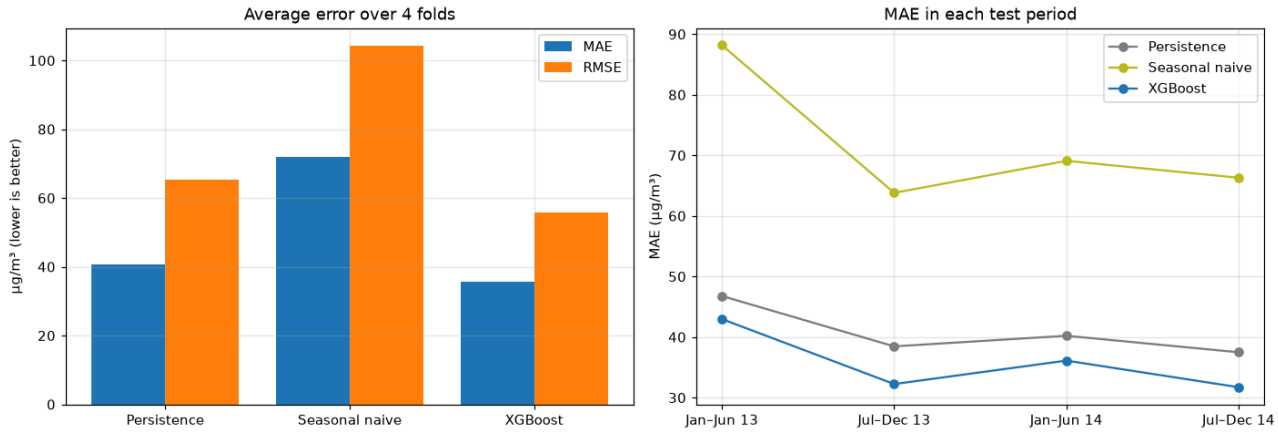
Fold	Training data	Test period	Test hours
1	Jan 2010 – Dec 2012	Jan – Jun 2013	4,309
2	Jan 2010 – Jun 2013	Jul – Dec 2013	4,369
3	Jan 2010 – Dec 2013	Jan – Jun 2014	4,311
4	Jan 2010 – Jun 2014	Jul – Dec 2014	4,344

As a **leakage guard**, the last 6 training hours (whose targets fall inside the test period) are kept as history but not used as targets. All models are scored on the same hours with **MAE** (average error size), **RMSE** (penalises large misses) and **R^2** (share of variance explained), and the 4 folds are averaged.

4. Results

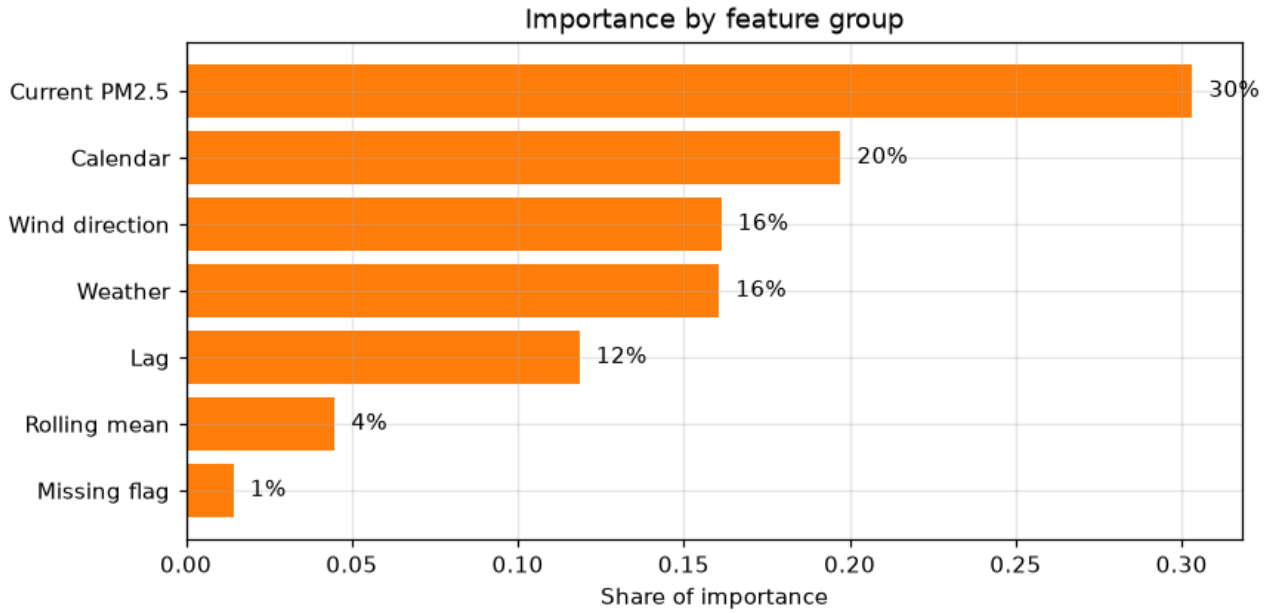
XGBoost is the best model in every fold. Averaged over the 4 test periods (Jan 2013 to Dec 2014):

Model	MAE (µg/m³)	RMSE (µg/m³)	R^2	AQI category accuracy
Persistence	40.73	65.31	0.518	49.8%
Seasonal naive	71.84	104.19	−0.223	31.5%
XGBoost	**35.76**	**55.80**	**0.652**	**53.9%**



- XGBoost lowers MAE by **12% on average** compared with persistence, and by 8% to 16% in individual folds.
- **Seasonal naive** performs worst, with a negative R^2 (worse than predicting the average). Yesterday's level is a poor guide 6 hours ahead, because pollution episodes build and clear within a day or two.
- **Error pattern:** XGBoost's errors are largest in January and February (MAE about 62 and 50), during the heating season and sudden spikes. They are smallest in summer (about 24 to 27).
- **AQI categories:** we map each forecast to the US EPA 2024 categories (Good ≤ 9.0, Moderate ≤ 35.4, Unhealthy for Sensitive Groups ≤ 55.4, Unhealthy ≤ 125.4, Very Unhealthy ≤ 225.4, Hazardous above). XGBoost picks the correct category for 53.9% of hours, compared with 49.8% for persistence. EPA defines these breakpoints on 24-hour averages, so applying them to hourly forecasts is a simplification.

What the model relies on. In the final fold, current PM2.5 carries about 30% of XGBoost's feature importance, calendar features 20%, wind direction and weather about 16% each, and the hand-picked lags only 12%. The single most useful weather signal is northwest wind.



All 14 project figures (EDA, backtest, forecasts, errors, importance) are in the repository under `reports/figures/`.

5. Discussion, Limitations and Future Work

Why XGBoost wins. Persistence already captures most of the short-term signal, because PM2.5 changes slowly. XGBoost adds value by learning *when* pollution is likely to rise or fall: at night, in winter, under calm air or before a northwest wind clears it. A simple rule cannot capture these combined effects.

Limitations

- **Sudden spikes are hard to predict.** Some extreme episodes (above 400 µg/m³) start within a few hours, and there are no early signs of them in 6 hours of history.
- **Fixed, hand-picked windows.** Lags only reach 24 hours, and the model gives them limited weight. Longer patterns, such as the build-up over several stagnant days, are only partly captured.
- **Single station and period.** The results apply to one Beijing monitor from 2010 to 2014. Emissions and regulations have changed since then.
- **Simplified AQI.** We apply 24-hour EPA breakpoints to hourly values.
- **No weather forecasts.** We only use observed weather. In practice, forecast wind and rain for the next 6 hours would likely help a lot.

Future work

1. Train an **LSTM** sequence model on the last 48 hours of raw data to test whether learned temporal patterns beat hand-picked lags.
2. Add **weather forecast** inputs and pollution data from nearby stations.
3. Tune hyperparameters with a time-ordered validation split, and evaluate several horizons (1, 6, 12 and 24 hours).
4. Forecast 24-hour averages directly for a fully standard AQI.

6. Conclusion

Under a fair rolling chronological backtest, an XGBoost model with lag, rolling, calendar and weather features forecasts 6-hour-ahead PM2.5 in Beijing **12% more accurately than persistence** (MAE 35.8 vs 40.7 µg/m³). It beats both baselines in every test period. The pipeline is reproducible from one public repository: 6 notebooks, 14 figures and 10 automated tests regenerate every number in this report.

Team contributions

Member	Responsibility	Files
Manan Gilhotra	Project setup, data loading and cleaning	config.py, data_loader.py
Harshita	Feature engineering and target	features.py
Satyam	Rolling backtest, metrics and baselines	backtest.py, baselines.py
Yashi Agarwal	XGBoost model	xgboost_model.py
Akhil Mishra	AQI mapping, charts, pipeline runner, dashboard	aqi.py, plots.py, main.py, app.py

References

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3. Chen, T. and Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. *Proceedings of KDD 2016*.
4. Hochreiter, S. and Schmidhuber, J. (1997). Long Short-Term Memory. *Neural Computation*, 9(8).
5. US EPA (2024). *Final Updates to the Air Quality Index for Particulate Matter*.